

# Capsular fragility, zonules and IOL support

Part 2 | Zonular failure has preoperative and intraoperative clues

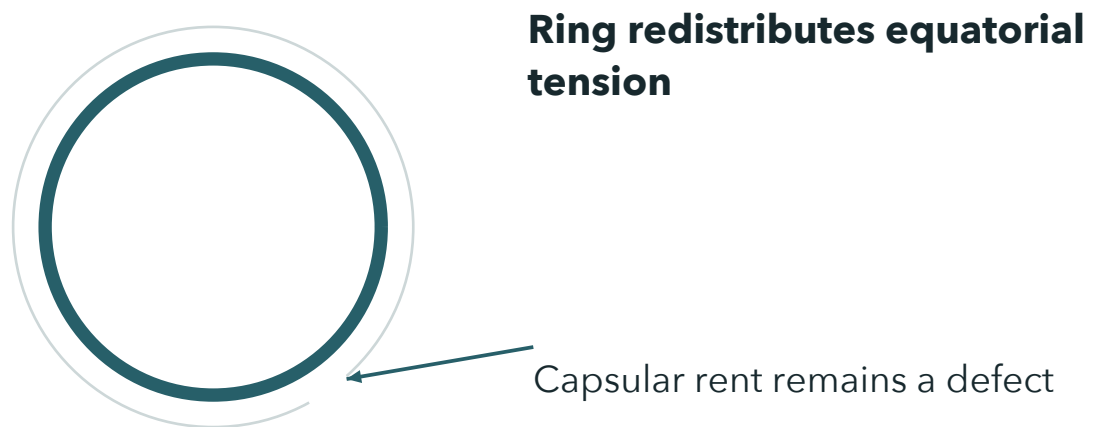
A small pupil and a mobile lens can signal more than one risk.

| Clue                               | Interpretation                              |
|------------------------------------|---------------------------------------------|
| Phacodonesis or asymmetric chamber | Assess lens support                         |
| Capsule moves during rhexis        | Possible zonular instability                |
| Progressive dialysis               | Support strategy may need escalation        |
| Trauma or systemic phenotype       | Changes the expected extent and progression |

Anticipate the need for capsular support and alternatives if the bag cannot safely hold an implant. Apparent absence of preoperative movement does not exclude weakness. Intraoperative behaviour should change the plan before force is repeatedly transmitted to an unstable capsule.

# A ring redistributes tension in an intact bag

It is not a repair for a capsular tear.



Bag shown in plan view

Original plan view: an equatorial ring distributes support; a rent remains a discontinuity.

A standard capsular tension ring can help selected zonular weakness when capsular integrity is suitable. Progressive or extensive loss may need fixation devices or another IOL strategy. A posterior capsular rent changes suitability because insertion or expansion can extend the defect or displace the device; the ring does not recreate a posterior barrier.

# Posterior polar cataract changes fluid and rotation choices

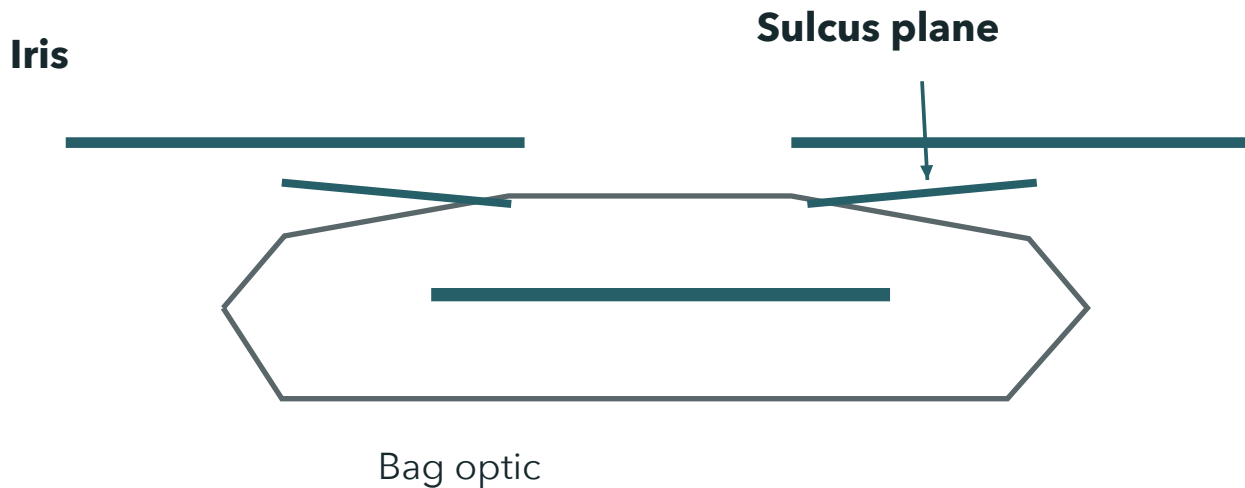
A vulnerable posterior capsule should be treated as a planning constraint.

| Routine action           | Reason for caution                                 |
|--------------------------|----------------------------------------------------|
| Forceful hydrodissection | Can stress a pre-existing weak posterior area      |
| Nuclear rotation         | Can transmit force to an adherent posterior plaque |
| Abrupt chamber change    | May expose or extend a defect                      |
| Plaque removal           | May require a staged decision                      |

Use a technique that respects posterior capsular fragility, including controlled lens separation and fluid delivery. Exact methods depend on anatomy and surgeon experience. Recognizing the pattern before surgery permits contingency planning for vitreous management and IOL support, rather than discovering the risk after a routine maneuver causes rupture.

# Implant position determines which design is suitable

The bag, sulcus, iris and sclera are different support environments.



Original section: the sulcus lies behind iris and anterior to the capsular bag.

When capsular support is inadequate, choose a fixation method from remaining tissue, corneal status, anatomy and surgical expertise. A conventional single-piece acrylic bag IOL is generally unsuitable for sulcus placement because its haptics can chafe the iris. A suitable alternative is not chosen by spherical power alone.

# Scleral fixation introduces its own tradeoffs

Secure centration and tissue compatibility require a long-term plan.

| Choice or risk                 | Reasoning consequence                         |
|--------------------------------|-----------------------------------------------|
| Suture or sutureless technique | Different fixation and failure mechanisms     |
| Optic tilt or decentration     | Can degrade visual quality                    |
| Haptic or suture exposure      | Needs surveillance and management             |
| Possible future gas or air     | Consider material-related opacification risks |

No secondary-fixation method is best for every eye. Consider available iris tissue, corneal endothelium and posterior-segment needs. Discuss the possibility of later instability, infection or optical symptoms. Technique selection belongs to a suitably experienced surgeon and should include rescue options if fixation is unsatisfactory.